

Second Latitude: A Futarchy-Based Funding Protocol

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1 Abstract

Second Latitude is a futarchy-based funding protocol for web3 tokenomics. Using prediction markets and community governance, it allocates capital to projects during periods of significant market uncertainty.

2 Introduction

Fundraising in web3 is broken. Builders face a flood of uncertainty—investors hesitate without strong foundational signals, and capital often flows based on hype rather than substance.

Second Latitude offers a new model—one grounded in market-driven confidence. We combine prediction markets with community governance to fund builders not just on belief, but on collective conviction.

Our futarchy-based system allows investors to stake **for** or **against** a project’s success. Believers back projects with 2LAT tokens, while skeptics can “short” them—profiting if their predictions are correct.

3 Token Architecture

Our platform operates with two distinct tokens:

2LAT Token

- Fairlaunched on Solana via a bonding curve (e.g., `pump.fun`).
- Total supply: 1,000,000,000 tokens (no additional issuance).
- Purpose: Facilitates cycle access payments and staking in prediction markets.

MOON Token

- Non-transferable token issued upon cycle payment.
- Purpose: Enables voting on projects entering prediction markets.
- Lifecycle: Burns at cycle’s end if unused, ensuring scarcity.

4 Moon Cycles and Access

- **Cycle Duration:** One moon cycle lasts 29.5 days.
- **Access Payment:**
 - Users pay a fixed amount A in 2LAT to join a cycle.
 - Breakdown:
 - * **Locked Amount:** $L_i = \frac{9}{10}A$, locked for 3 cycles.
 - * **Treasury Allocation:** $T_i = \frac{1}{10}A$, used for builder subsidies.
 - Reward: Users receive M MOON tokens per payment for governance voting.
- **Locking Mechanism:**
 - For cycle i , the locked amount is $L_i = \frac{9}{10}P$.
 - Unlocks occur after 3 cycles: L_{i-3} becomes available.
 - Skipping a cycle burns all prior locked amounts.
 - **Benefit:** Long-term participants can reduce their effective cost by up to 90%.

5 Project Selection via Governance

- Users nominate or vote for builders by allocating MOON tokens.
- Each builder B_i accumulates votes $m_{i,j}$ from user j .
- Builders reaching the threshold $V_i = \sum_j m_{i,j} \geq \theta$ become eligible to offer a prediction market for their token and receive Builder Subsidies, where θ is a constant throughout the cycle.

6 Builder Subsidies

Treasury and Subsidy Pool

- 10% of each cycle payment contributes to the treasury.
- The cumulative treasury across users forms the subsidy pool:

$$\mathcal{R} = \sum_{i=1}^N T_i = \sum_{i=1}^N \frac{1}{10} A_i$$

- This pool is allocated to builders via quadratic funding.

Quadratic Subsidy Distribution with Threshold

Builders receive subsidies based on the square of the sum of square roots of votes, encouraging broader support:

$$Q_i = \begin{cases} \frac{\left(\sum_j \sqrt{m_{i,j}}\right)^2}{\sum_{k \in \mathcal{E}} \left(\sum_j \sqrt{m_{k,j}}\right)^2} \cdot \mathcal{R}, & \text{if } V_i \geq \theta \\ 0, & \text{if } V_i < \theta \end{cases}$$

- Q_i : Subsidy allocated to builder B_i .
- $\mathcal{E}$: Set of all eligible builders such that $B_k \in \mathcal{E} \iff V_k \geq \theta$.
- $m_{i,j}$: MOON votes from user j to builder B_i .
- $\mathcal{R}$: Total treasury reward pool from 10% of cycle access payments.

7 Futarchy-Based Prediction Markets

Let the following variables define the prediction mechanics:

- $\mathcal{S}$: Total 2LAT staked on success.
- $\mathcal{F}$: Total 2LAT staked on failure.
- $\mathcal{T} = \mathcal{S} + \mathcal{F}$: Total prediction market pool.
- $\phi = 0.05 \times \mathcal{T}$: Protocol fee.
- p_E : Price at which project tokens are entered on the prediction market (entering price).
- p_S : Expected or predicted project token price (market price), with $p_S > p_E$.
- p_F : Token price in case of failure, with $p_F < p_S$.
- $\delta = \min(p_S - p_E, 0.2)$: Project reward multiplier cap.
- $\mathcal{I} = \delta \times \mathcal{T}$: Project's immediate funding in 2LAT (project cut).
- $\mathcal{D} = \mathcal{T} - \phi - \mathcal{I}$: Distributable pool for predictors.

Project Funding

- The project receives $\mathcal{I}$ in 2LAT for development.
- Protocol uses $\mathcal{I}$ to buy project tokens at price p_E and distributes them to correct predictors.

Payouts

If success:

$$\begin{aligned}\text{Total payout to success predictors} &= \mathcal{D} + \left(\frac{\mathcal{I}}{p_E} \cdot p_S \right) \\ \text{Payout per user with stake } s_i &= \left(\frac{s_i}{\mathcal{S}} \right) \left[\mathcal{D} + \left(\frac{\mathcal{I}}{p_C} \cdot p_D \right) \right]\end{aligned}$$

If failure:

$$\begin{aligned}\text{Total payout to failure predictors} &= \mathcal{D} + \left(\frac{\mathcal{I}}{p_E} \cdot p_F \right) \\ \text{Payout per user with stake } f_i &= \left(\frac{f_i}{\mathcal{F}} \right) \left[\mathcal{D} + \left(\frac{\mathcal{I}}{p_C} \cdot p_F \right) \right]\end{aligned}$$

This design rewards correct predictions with upside exposure and ensures projects receive meaningful capital.

8 Builder Rewards

- **Direct Funding:** Projects receive $\mathcal{I}$ in 2LAT.
- **Token Incentives:** Tokens issued to the protocol are redistributed to correct predictors.
- **Community Boost:** MOON vote counts signal trust and drive visibility.

9 Cycle Summary

1. **Join:** Pay A in 2LAT (90% locked, 10% to treasury), receive M MOON.
2. **Vote:** Use MOON to select projects.
3. **Stake:** Place 2LAT on project success ($\mathcal{S}$) or failure ($\mathcal{F}$).
4. **Resolve:** Winners split $\mathcal{D}$ and project tokens; project receives $\mathcal{R}$.
5. **Reset:** MOON burns if unused; new cycle begins.

10 Conclusion

Second Latitude combines the rhythmic trust of 2.lat with futarchy's market-driven precision, creating a funding engine for builders who deliver and investors who discern value. We're raising 2LAT now to fuel this vision—join us to lead the future of web3 funding.